

FUNCTIONAL REQUIREMENTS DOCUMENT (FRD)

AI-Powered CMO Response Time Reduction & Incident Management Automation System

Defines the functional behavior, business rules, controls and non-functional requirements for the incident-management automation system.

Document Owner	Status	Version	Classification
Business Analyst / Project Lead	Portfolio-ready	1.0	Sanitized public case study

1. Purpose and Scope

This Functional Requirements Document translates approved business requirements into detailed system behavior. It covers the two solution modules: Reactive Incident Automation and Proactive Incident Monitoring. It is written for business users, delivery stakeholders and technical implementers using sanitized terminology.

2. Solution Overview

The solution receives incident updates from approved operational communication channels, classifies relevant alerts, creates structured notifications, identifies stakeholders using an approved mapping, schedules follow-ups, escalates long-running incidents, records processing activity and presents operational status in a dashboard. The solution supports - but does not replace - human incident resolution and decision-making.

3. Functional Modules

Module	Purpose	Primary Output
Input and Alert Processing	Receive approved incident updates and normalize usable fields.	Validated incident record or exception log
Incident Classification	Determine alert category, severity and relevance using configured rules.	Criticality and routing decision
Notification and Ownership	Create structured alerts and route to mapped stakeholders.	Owner notification
Follow-up and Escalation	Monitor open incidents and apply defined reminder and escalation thresholds.	Reminder or escalation notification
Restoration and Closure	Process valid restoration updates and update incident lifecycle status.	Closure notification and final status
Proactive Monitoring and Dashboard	Aggregate open critical incidents and performance measures.	Area-level view and executive KPI dashboard
Audit and Exception Management	Capture processing, failures and manual-review cases.	Traceable operational logs

4. Functional Requirements

ID	Functional Requirement	Related BRD	Priority
FR-01	The system shall capture incident messages from configured, approved operational sources.	BR-01	High
FR-02	The system shall extract available incident fields such as timestamp, location/area, incident reference, status and free-text description.	BR-01	High
FR-03	The system shall validate mandatory fields required for automated routing and log incomplete alerts for manual review.	BR-01, BR-09	High
FR-04	The system shall classify alerts using configured rules to identify critical or service-impacting incidents.	BR-02	High
FR-05	The system shall suppress duplicate notifications for matching alerts within the configured matching window.	BR-03	High
FR-06	The system shall generate a structured notification containing the available incident reference, area, severity, status and timestamp.	BR-03	High
FR-07	The system shall map a valid incident area/category to the configured response owner or group.	BR-03	High
FR-08	The system shall send an automated follow-up notification for eligible open incidents at the configured interval.	BR-04	High
FR-09	The system shall escalate eligible incidents when the configured duration and severity threshold is reached.	BR-05	High
FR-10	The system shall process a valid restoration update and change the incident lifecycle status to closed/restored.	BR-06	High
FR-11	The system shall publish a view of pending critical incidents by configured area and status.	BR-07	High
FR-12	The system shall display agreed operational KPIs, including CMO-related docket trends and duration bands.	BR-08	Medium
FR-13	The system shall log processing outcomes, notification attempts, exceptions and manual-review events.	BR-09	Medium
FR-14	The system shall keep credentials and personal or company identifiers outside public code and portfolio documentation.	BR-10	High
FR-15	The system shall generate a health or failure indication when a processing error prevents completion.	BR-09, BR-11	Medium
FR-16	The system shall allow approved configuration changes to routing, thresholds and parsing rules without exposing sensitive values.	BR-12	Medium

5. End-to-End Workflow

Step	System Behavior	Outcome
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Step	System Behavior	Outcome
1. Receive	Receive an approved incident update.	Raw input is available for processing.
2. Validate	Extract fields and check required information.	Valid record or exception requiring manual review.
3. Classify	Apply configured rules to identify relevance and criticality.	Routing and priority decision.
4. Notify	Create structured alert and send to mapped owner(s).	Timely ownership visibility.
5. Track	Monitor the open record and issue reminders.	Consistent follow-up.
6. Escalate	Apply configured threshold rules for eligible incidents.	Escalation to appropriate authority.
7. Close	Process valid restoration update and send closure notification.	Lifecycle status updated.
8. Report	Refresh monitoring view and KPI dashboard.	Current operational and leadership visibility.

6. Detailed Business Rules

- Routing requires a recognized incident category and a valid owner mapping.
- The system must not send duplicate notifications for the same matching incident within the configured matching window.
- An alert that fails mandatory-field validation must be logged and flagged for manual review, not silently dropped.
- Follow-up is sent only for open, eligible incidents.
- Escalation is triggered only when both approved eligibility criteria and threshold conditions are met.
- Restoration updates close an incident only when they match an open record or contain the required valid reference.
- Dashboard metrics must show their refresh timestamp and should distinguish open from restored incidents.

7. Data Requirements

Data Element	Purpose	Handling Requirement
Incident Reference	Uniquely identify and match incident updates.	Required for reliable lifecycle tracking where available.
Timestamp	Measure sequence and duration.	Store with source time and processing time where available.
Area / Location	Route to owner and group dashboard results.	Use configured area mapping; sanitize in public examples.
Severity / Criticality	Apply prioritization and escalation rules.	Derived from configured rules or approved source field.
Status	Determine open, pending, restored or	Validate against allowed status values.

Data Element	Purpose	Handling Requirement
	exception state.	
Owner Mapping	Identify responsible team or role.	Maintain as restricted configuration; never publish real group IDs.
Processing Log	Provide auditability and troubleshooting.	Record outcome, error class and timestamp without exposing credentials.

8. Notification Logic

Event	Recipient	Required Notification Content
New critical incident	Mapped response owner(s)	Incident reference, area, severity, status, received time and required action.
Follow-up interval reached	Current response owner(s)	Open incident reference, elapsed duration and request for status update.
Escalation threshold reached	Configured escalation authority	Incident details, duration, outstanding status and escalation reason.
Valid restoration update	Relevant stakeholders	Incident reference, restoration time and closure/restored status.
Processing exception	Automation support / designated reviewer	Error summary, source reference where available and manual-review requirement.

9. Error Handling and Manual Fallback

- Malformed or incomplete input: log the exception and route to the designated manual-review process.
- Source or communication-platform unavailability: monitor service health, retry when permitted and use the existing operational fallback process.
- Unknown owner mapping: prevent automatic routing, log the case and request mapping correction.
- Dashboard refresh issue: display the last successful refresh time and notify support when refresh is beyond the defined threshold.
- Unexpected processing failure: retain the original event reference where permitted, log the technical error and alert the support owner without exposing sensitive details.

10. Non-Functional Requirements

Category	Requirement
Performance	Eligible alerts should be processed and notified promptly after receipt; actual target thresholds will be agreed with operations.
Availability	The solution should be monitored during the approved operational window, with a documented fallback when unavailable.

Category	Requirement
Security	Credentials, group IDs, personal identifiers and internal addresses must be held in restricted configuration and excluded from public artefacts.
Auditability	System actions, exceptions and notification outcomes must be traceable through logs.
Maintainability	Parsing, routing and threshold rules should be configurable and reviewable after operational validation.
Usability	Notifications and dashboards should be concise, readable and actionable for operations users.
Data Quality	The solution should validate mandatory data and clearly flag records that require manual review.

11. Reporting and Dashboard Requirements

- Show pending critical incidents by area and current status.
- Show current open versus restored incident counts for the selected reporting period.
- Show CMO-related docket comparison, including baseline and outcome figures where evidence is available.
- Show duration-band trends and a visible report refresh timestamp.
- Support executive-level review through concise KPI summaries and drill-down-ready operational data, subject to approved data access.

12. Future Enhancements

- Confidence scoring and human review queues for ambiguous alert classifications.
- Integration with approved ticketing or service-management platforms.
- Predictive risk indicators for incidents likely to breach thresholds.
- Role-based dashboard views and automated daily/weekly summary reports.
- Additional data-quality checks and performance analytics.

13. FRD Acceptance

This FRD is the functional baseline for detailed test cases, the Requirement Traceability Matrix (RTM) and User Acceptance Testing (UAT). Requirements will be validated through documented test scenarios and stakeholder sign-off.

Approver Role	Name / Signature	Date
Project Sponsor / Operations Leadership	_____ —	_____
Project Lead	_____ —	_____
Primary Business Representative	_____ —	_____